

Measure of Central Tendency.

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A single value intended to represent the whole data is called average. As it lies almost at the centre of data, therefore it is also known as measure of central tendency. Average is also used to locate the Central value of data therefore it is also known as measure of location.

Types of Averages

1. Arithmetic Mean
2. Median
3. Mode
4. Geometric Mean
5. Harmonic Mean.

Qualities (Properties) of Good Average.

1. It should be simple to understand and easy to calculate.
2. It should be clearly defined by a mathematical formula.
3. It should be based on all the observations of the data.
4. It should not be affected by extreme values.
5. It should be Calculated in all the Cases.
6. It should be capable of further mathematical treatment
7. It gives equal weightage to all the values.

Arithmetic Mean: It is defined as sum of all the values of a variable and divided by the total number of items.

$$\bar{X} = \Sigma X / n$$

Weighted Mean: Arithmetic Mean Computed for different items by considering relative importance of each item is called Weighted Mean.

$$\bar{X}_w = \Sigma X_w / \Sigma w$$

Properties of Arithmetic Mean.

1. The Sum of deviations of the observations from their mean is equal to zero.
that is $\Sigma(X - \bar{X}) = 0$
2. The sum of squares of the deviations of the observations from their mean is minimum.
that is $\Sigma(X - \bar{X})^2$ is minimum
3. The arithmetic mean is affected by change of origin and scale. If we add or subtract a constant from all observations or multiply or divide all the observations by a constant then arithmetic mean is changed.
If $y = ax + b$ where x and y are variables and a, b are constants, then $\bar{y} = a\bar{x} + b$
4. If $n_1, n_2, n_3, \dots, n_k$ observations have means $\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_k$ respectively then combined mean of total observations is given by

$$\bar{X}_C = (n_1\bar{X}_1 + n_2\bar{X}_2 + n_3\bar{X}_3 + \dots + n_k\bar{X}_k) / (n_1 + n_2 + n_3 + \dots + n_k)$$

Advantages (Merits) of Arithmetic Mean

- I. It is simple to understand and easy to Calculate.
- II. It depends on all the values of data
- III. It is clearly defined by a mathematical formula
- IV. It is capable of further algebraic analysis
- V. It is least affected by fluctuation of sampling.

Disadvantages (Demerits) of Arithmetic Mean

- I. It is greatly affected by the extreme values.
- II. It cannot be calculated accurately for open-ended frequency distribution.
- III. It cannot be Calculated for qualitative data
- IV. It is not appropriate average for highly skewed distribution

Median. Median is the middle value of an arranged set of data. It divides an arranged set of data into two equal parts.

Advantages of Median

- I. It is simple to understand and easy to calculate.
- II. It is not affected by extreme values.
- III. It Can be Computed from an open-end frequency distribution.
- IV. It is suitable average for qualitative data
- V. It is an appropriate average in a highly skewed distribution
- VI. It can be located graphically.

Disadvantages of Median

- I. It is not rigidly defined by a mathematical formula.
- II. It is not based on all the values.
- III. It is necessary to arrange the data which is time consuming and difficult-task.
- IV. It is not Capable of further mathematical treatment.

Mode: Mode is that value which occurs the greatest number of times in the data.

Advantages of Mode

- I. It is simple to understand and easy to calculate
- II. It is not affected by extreme values.
- III. It Can be Computed from an open-end frequency distribution.
- IV. It is suitable average for qualitative data
- V. It can be located graphically

Disadvantages of Mode

- I. It is not rigidly defined by mathematical formula.
- II. It is not based on all the values.
- III. Sometimes a distribution may have no mode.

IV. It is not capable of further mathematical treatment .

Quartiles: Quartiles divide an arranged set of data into four equal parts.

Deciles: Deciles divide an arranged set of data into ten equal parts.

Percentiles: Percentiles divide an arranged set of data into hundred equal parts.

Geometric Mean: It is the nth root of the product of 'n' positive non zero values.

$$GM = [(X_1, X_2, \dots, X_n)]^{1/n}$$

Advantages of Geometric Mean

- I. It is defined by a mathematical formula.
- II. It depends on all the values of data.
- III. It gives equal weightage to all the values
- IV. It is appropriate average for calculating average percentage increase in population, income etc.

Disadvantages of Geometric Mean

- I. It is neither easy to calculate nor simple to understand.
- II. It cannot be Calculated in Case of zero or negative values in data

Harmonic Mean

It is defined as the reciprocal of the arithmetic mean of the reciprocal of the values

$$HM = n / (\sum 1/x)$$

Advantages of Harmonic Mean

- I. It is defined by a mathematical formula.
- II. It depends on all the values of data.
- III. It is not affected by fluctuation of sampling.
- IV. It is appropriate average for Calculating average Speed, average price etc

Disadvantages of Harmonic Mean

- I. It is neither easy to understand nor easy to Calculate.
- II. It cannot be calculated if any value in the data is zero.
- III. It is greatly affected by extremely small values.

Empirical Relationship.

In a symmetrical distribution

Mean = Median = Mode

In a moderately Skewed distribution

Mode = 3 Median - 2 Mean.